

AUTOMATED GLACIAL LAKE DETECTION USING DEEPLABV3+

Team Name: Trixel

Team Members:

1. Hariram J
2. Manoj Kumar S
3. Praveen A M

Mentor:

1. Dr.B.Sathya Bama

Institution: Thiagarajar College of Engineering

Abstract

Glacial lakes are rapidly increasing in number and size due to climate change and glacier retreat. These lakes pose a significant risk because they can trigger Glacial Lake Outburst Floods (GLOFs), causing severe damage to downstream communities and infrastructure. Manual mapping of glacial lakes from satellite imagery is labor-intensive and time-consuming. This project presents an automated deep learning framework for glacial lake detection using DeepLabV3+ with an EfficientNet-B4 backbone. The proposed system performs semantic segmentation on satellite imagery to accurately identify lake regions while minimizing confusion with snow, glacier ice, debris-covered ice, and terrain shadows. The model achieved an Accuracy of 99.62%, Precision of 98.09%, Recall of 93.49%, F1 Score of 95.74%, IoU of 91.82%, and Cohen's Kappa of 0.955.

Class-wise Output

1. Cloud Cover

Description: Output images obtained for cloud-covered regions

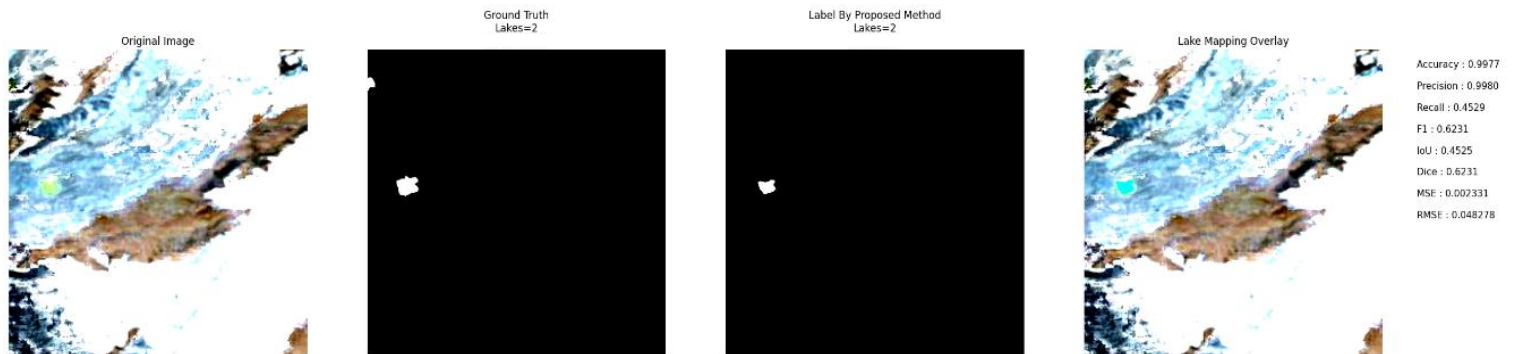


Image No: 10

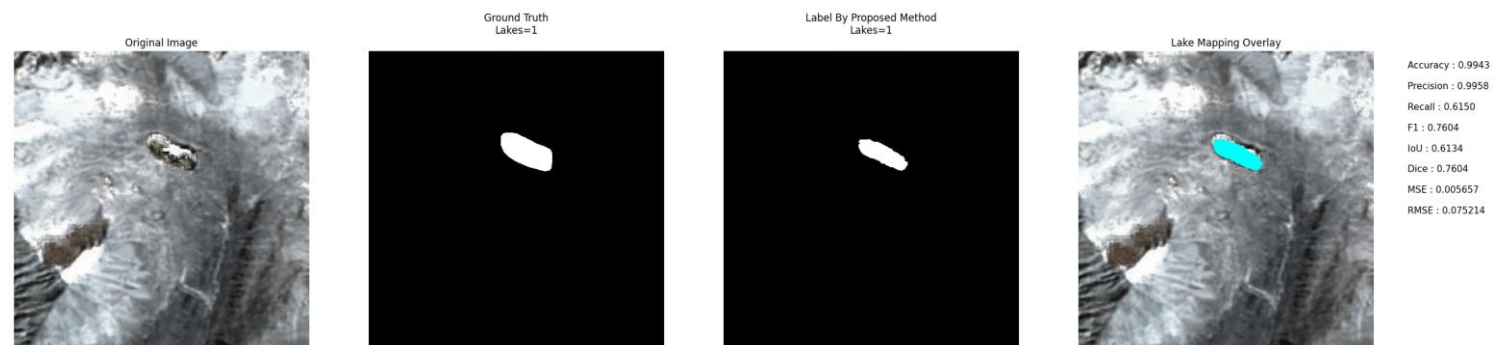


Image No: 168

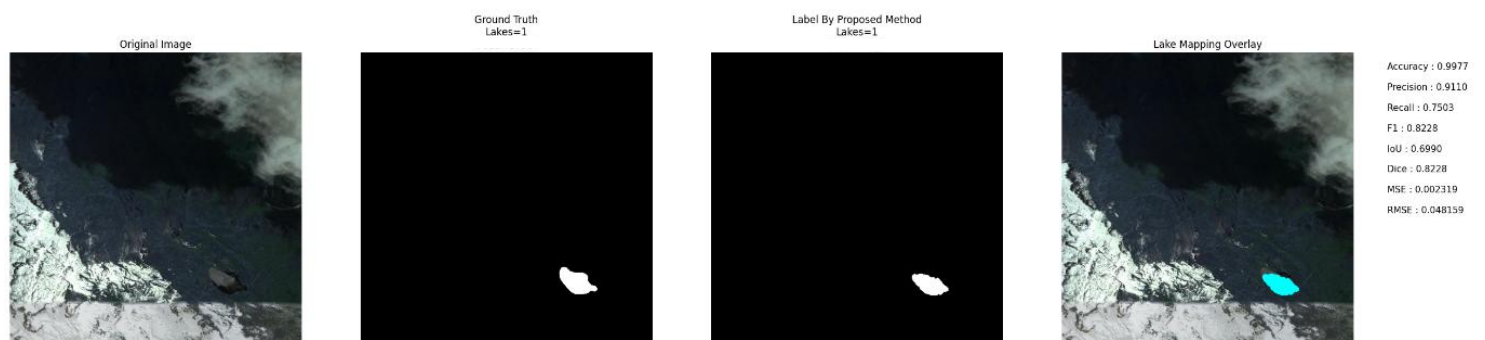


Image No: 192

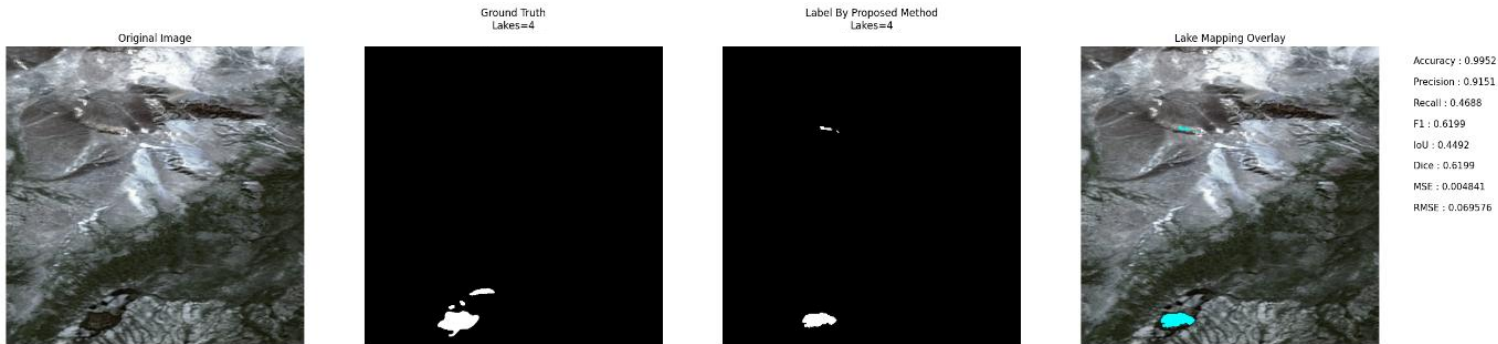


Image No: 320

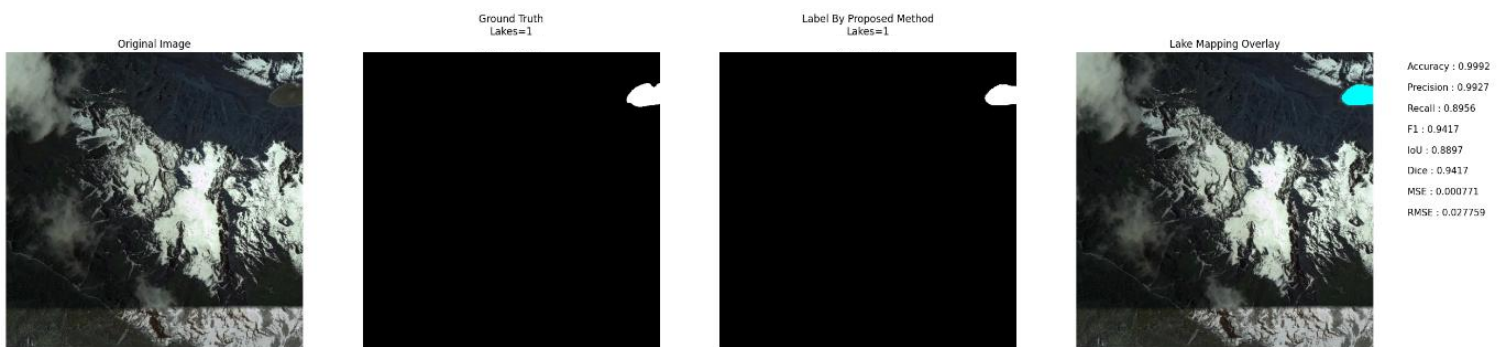


Image No: 427

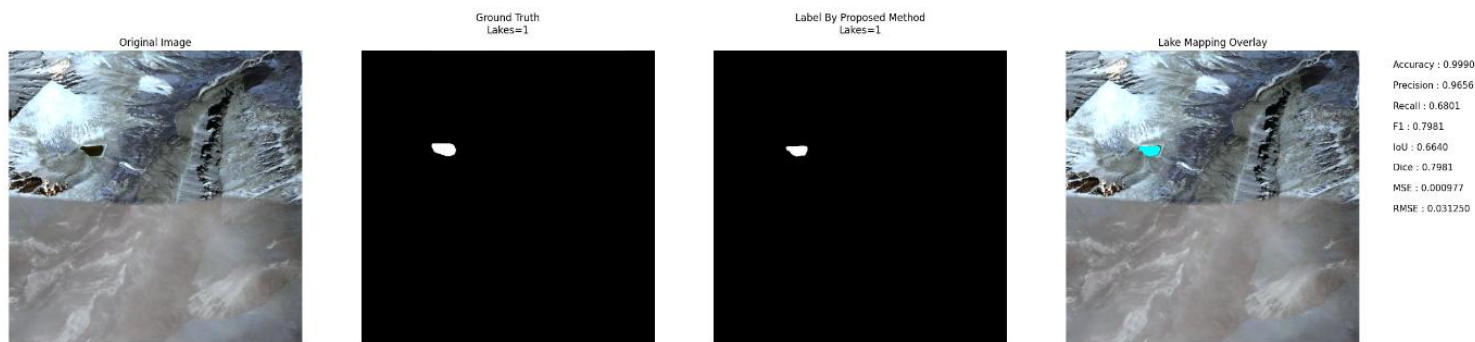


Image No: 488

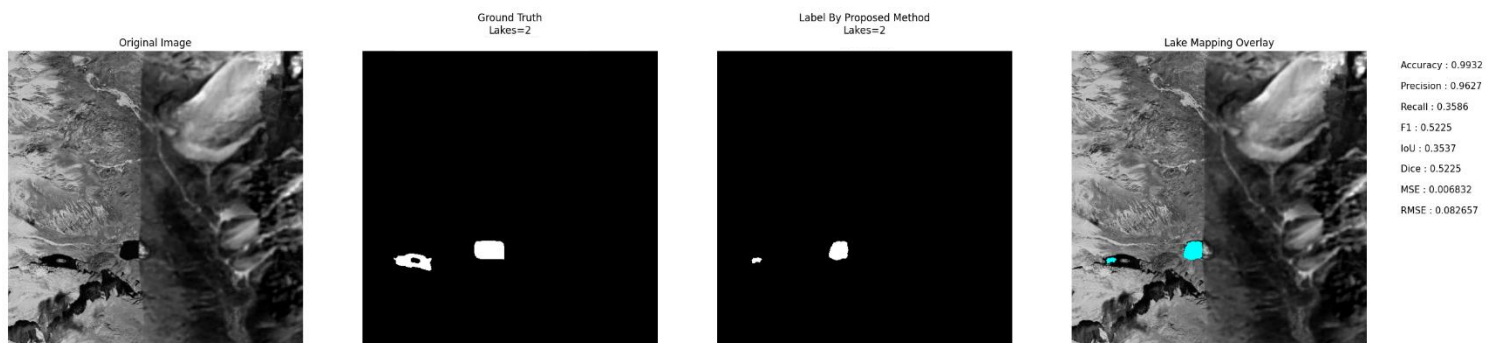


Image No: 588

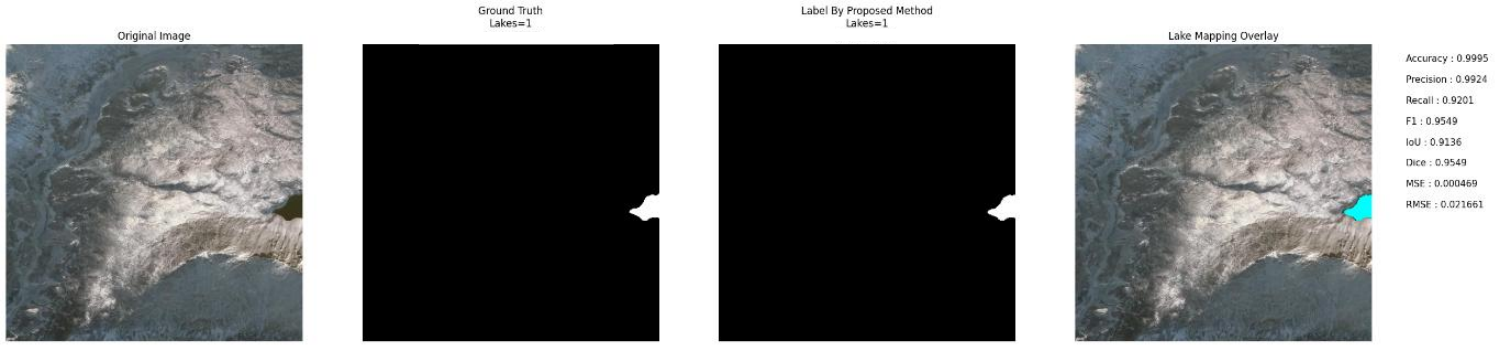


Image No: 686

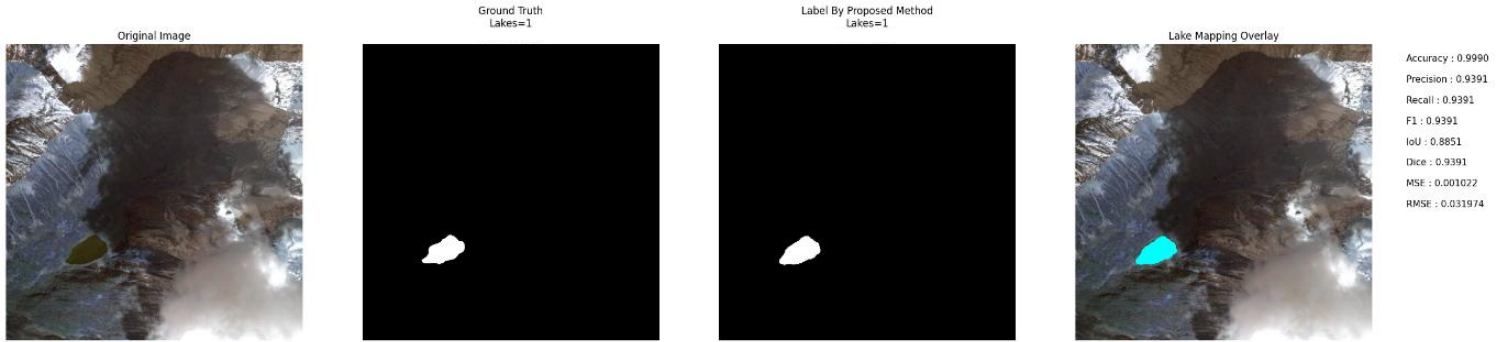


Image No: 725

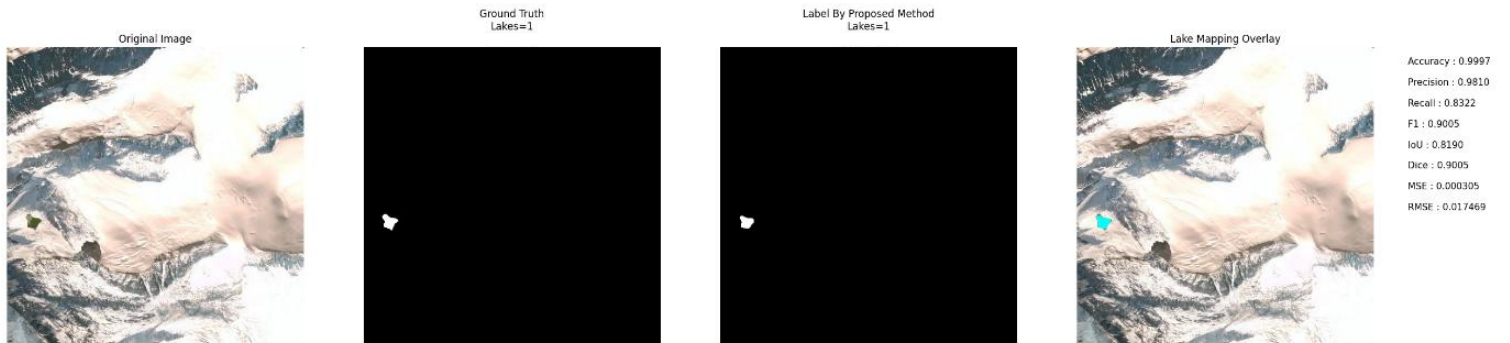


Image No: 74

2. Snow Cover

Description: Output images obtained for snow-covered regions.

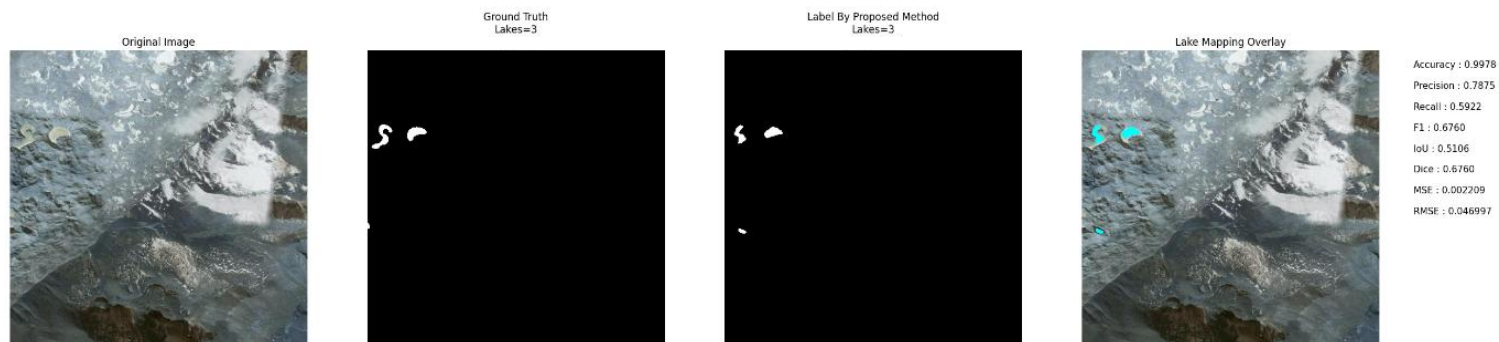


Image No: 102

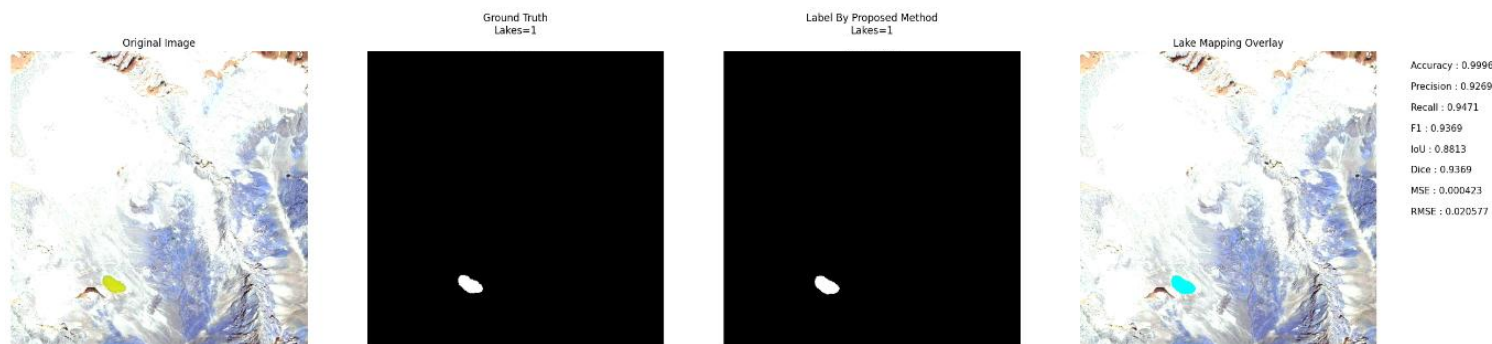


Image No: 1173

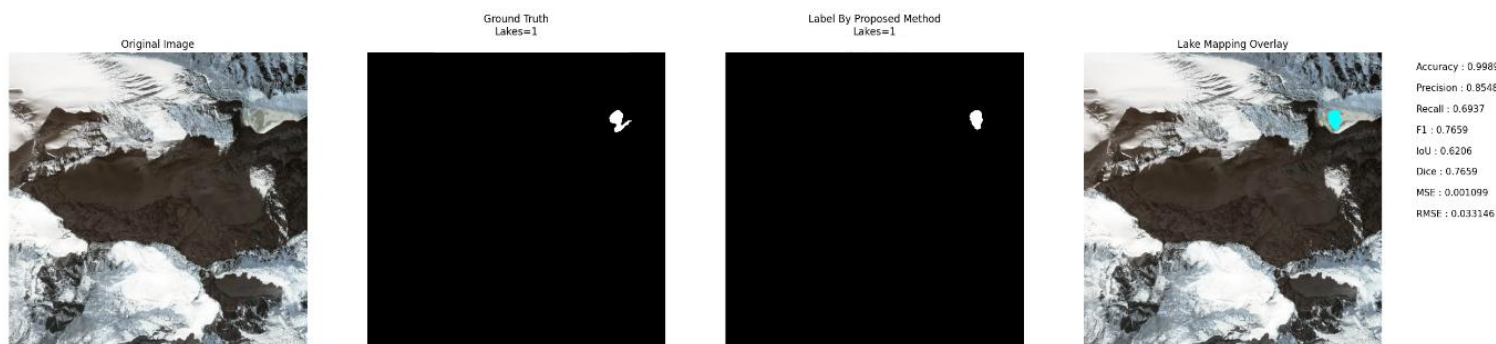


Image No: 108

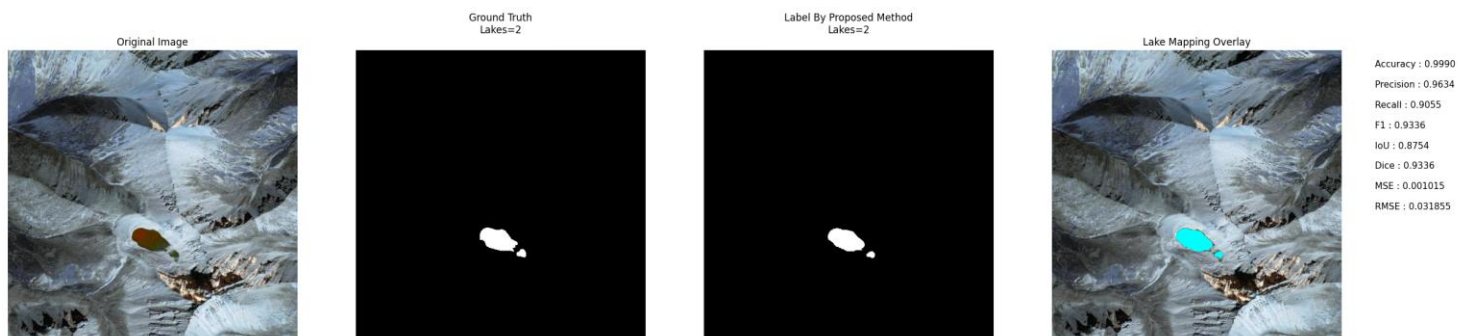


Image No: 117

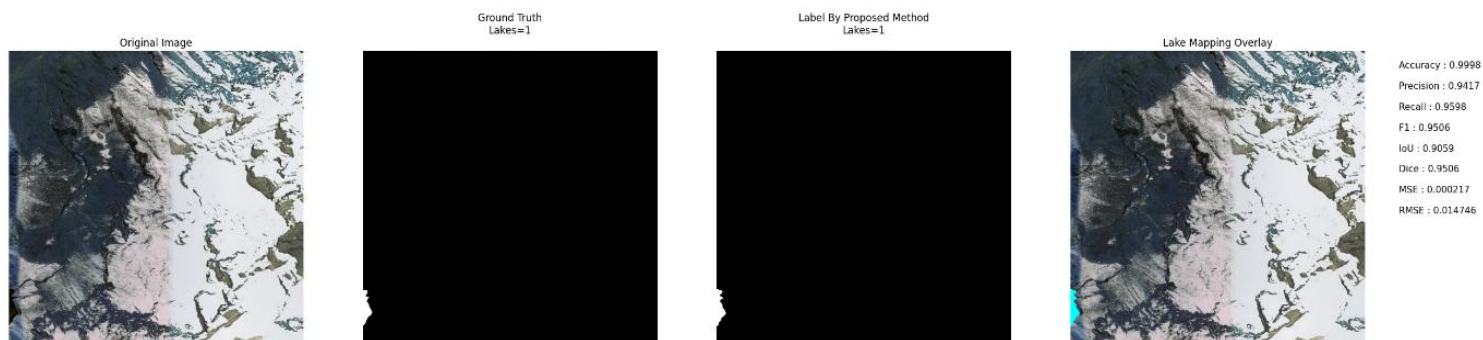


Image No: 1003

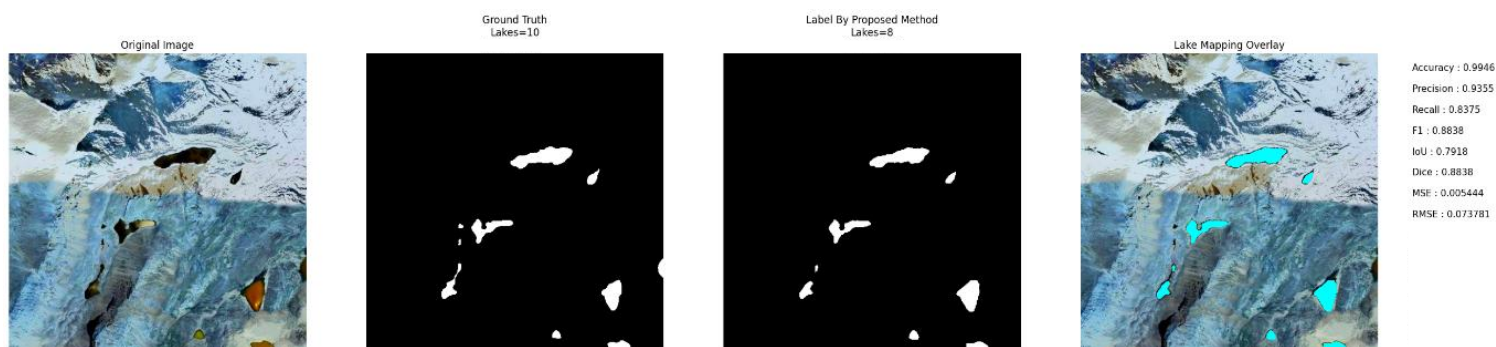


Image No: 1006

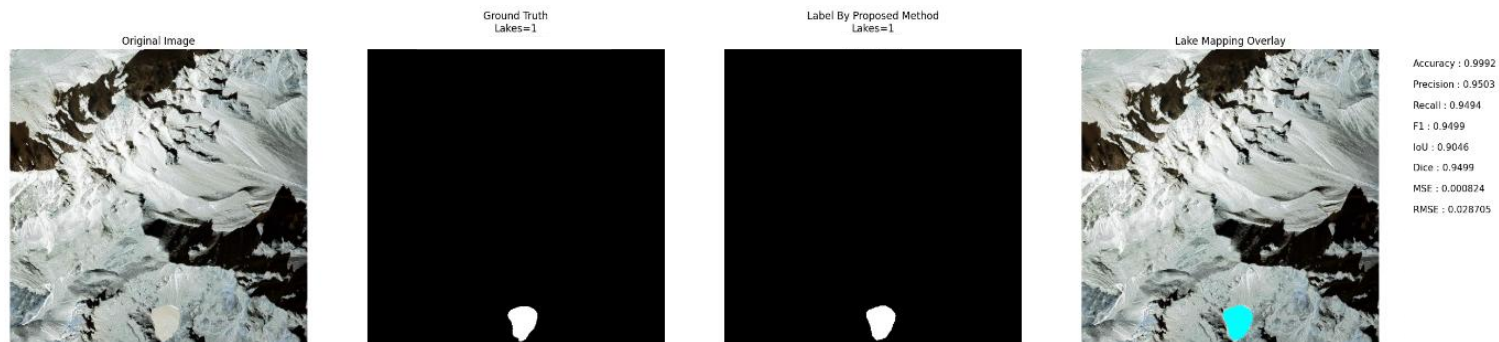


Image No: 1010

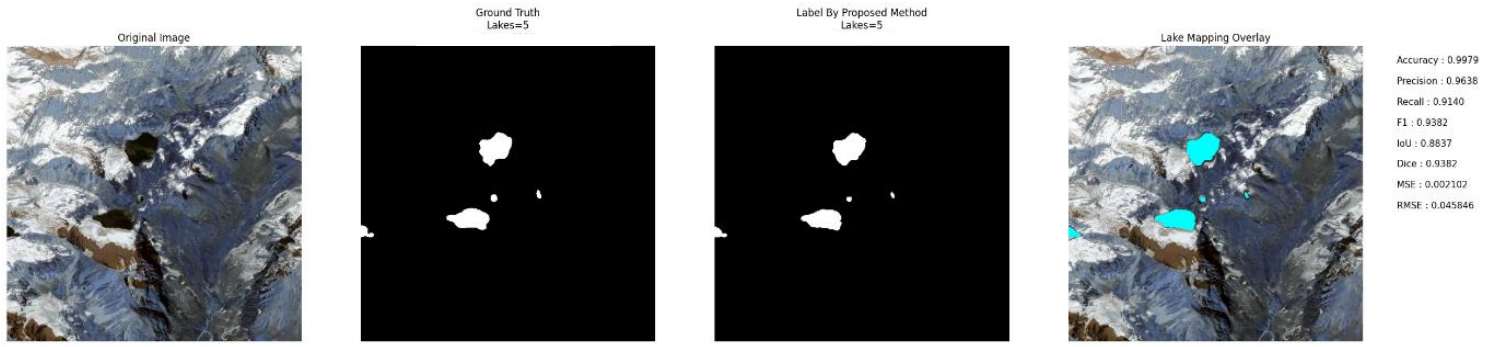


Image No: 1013

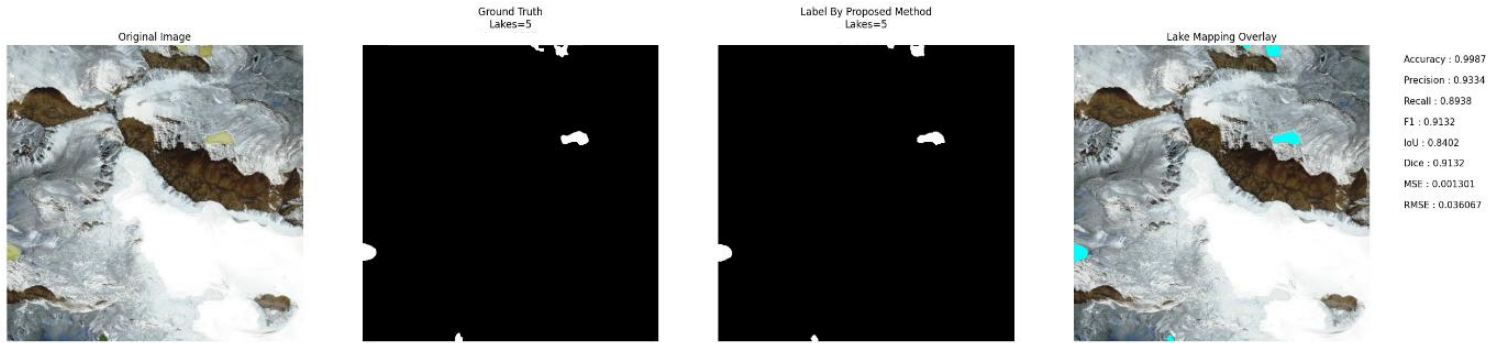


Image No: 1015

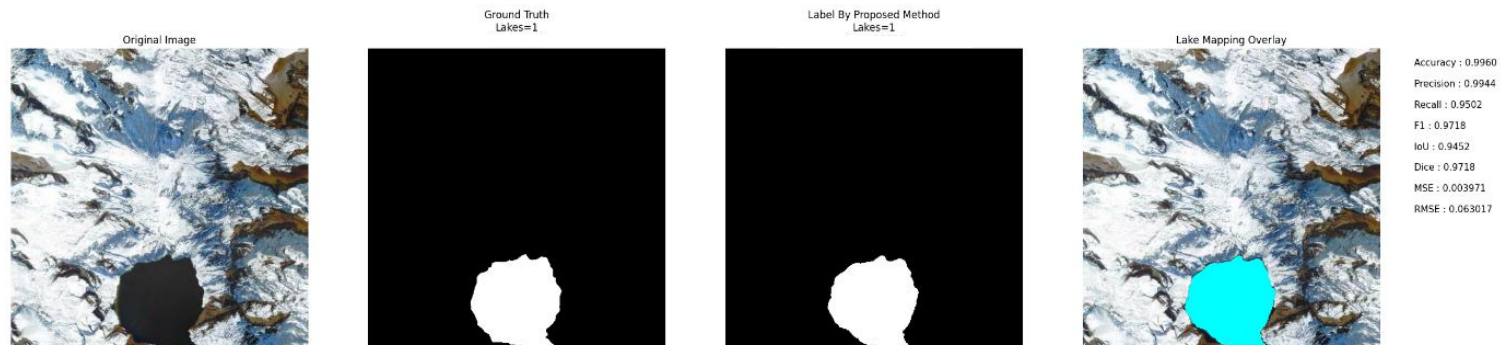


Image No: 1018

3. Terrain Shadow

Description: Output images obtained for terrain shadow regions.

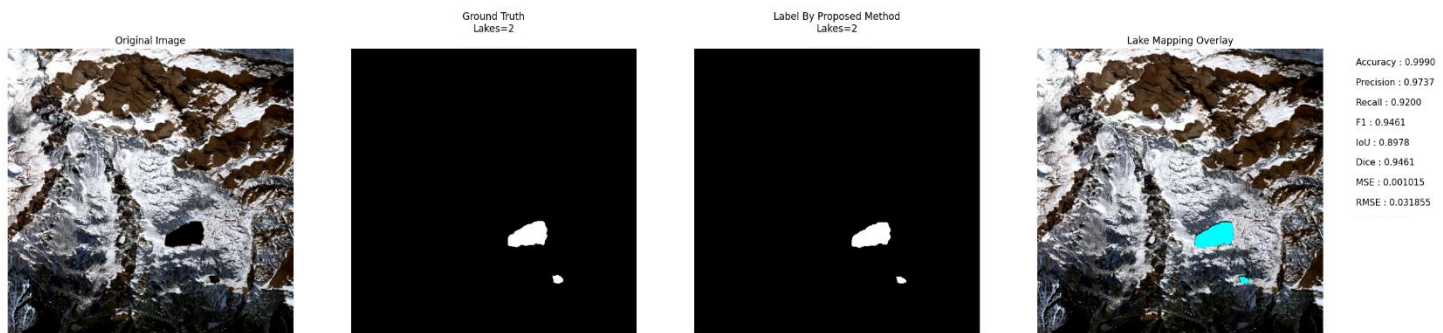


Image No: 524

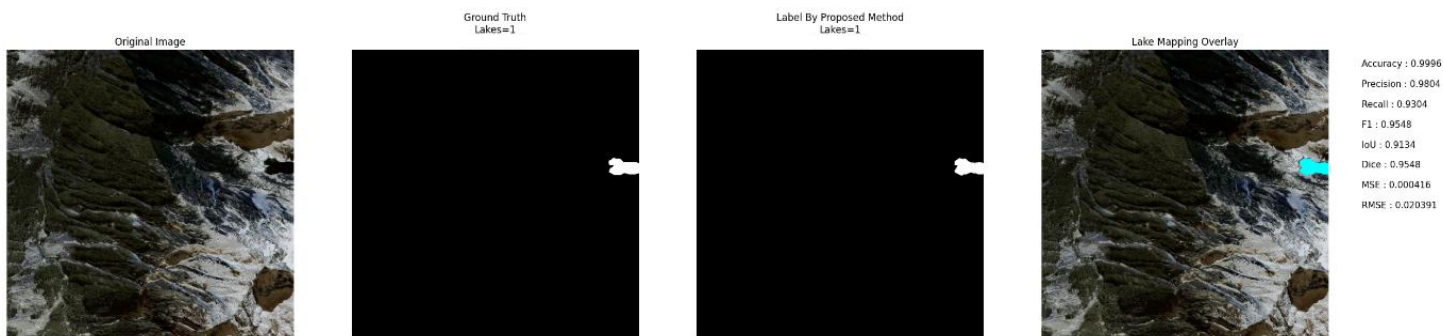


Image No: 1681

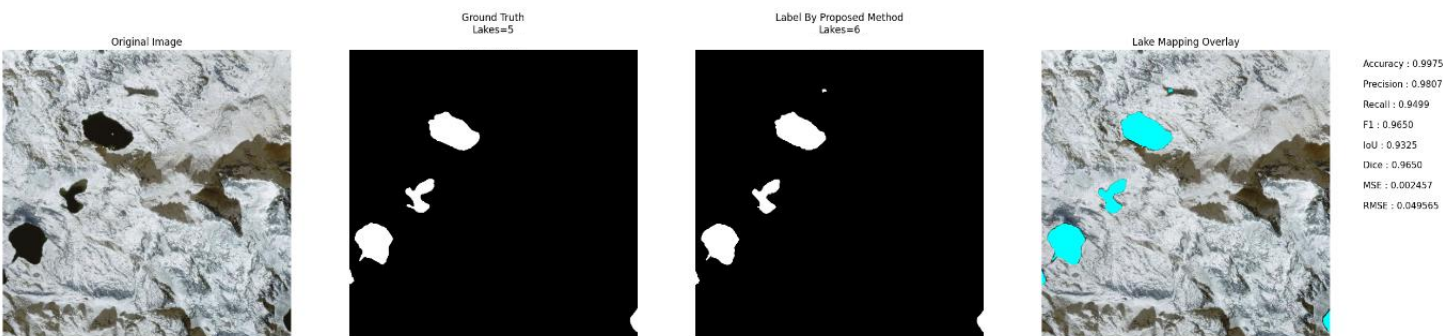


Image No: 180

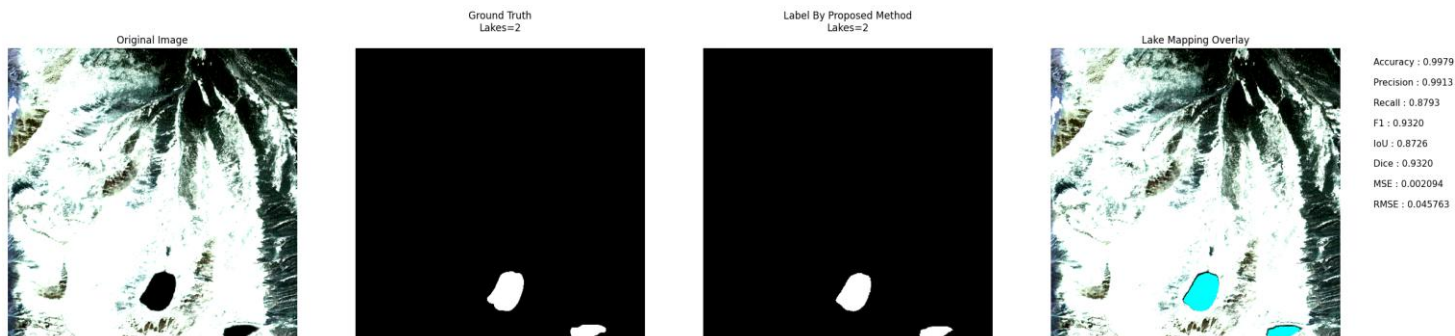


Image No: 457

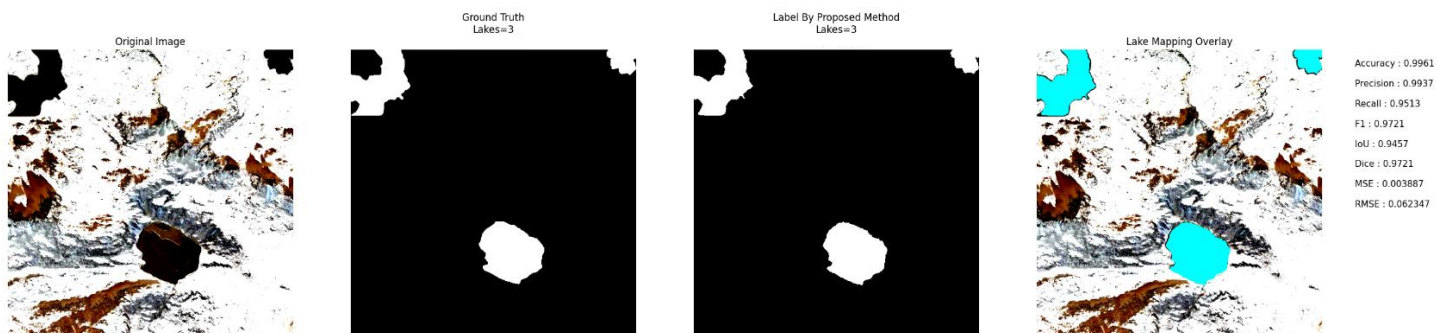


Image No: 263

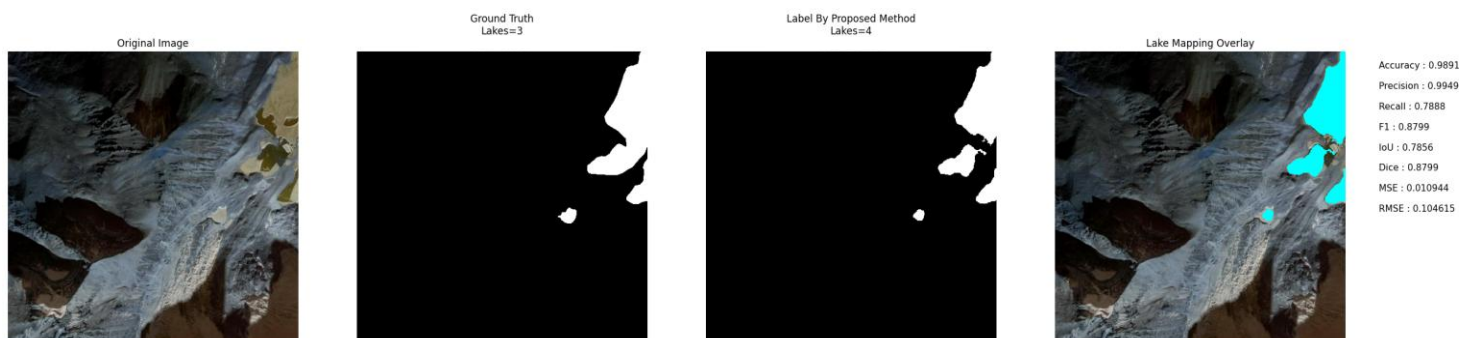


Image No: 530

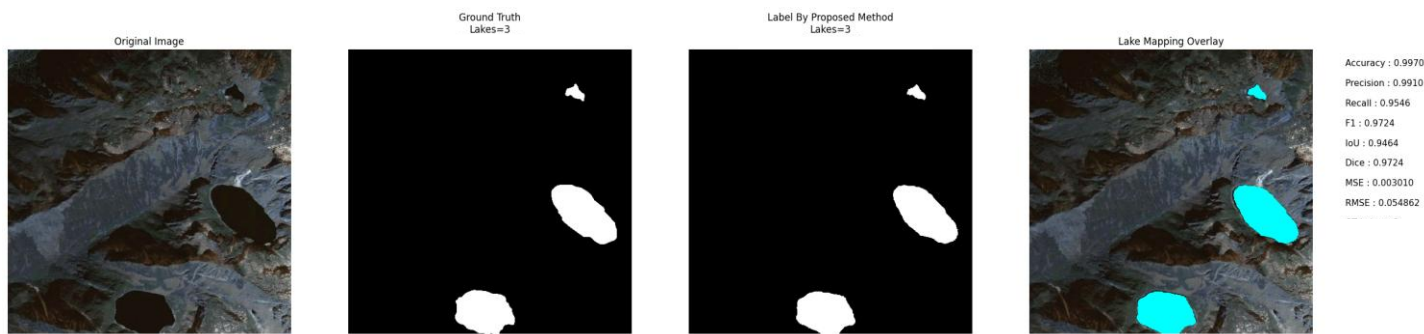


Image No: 309

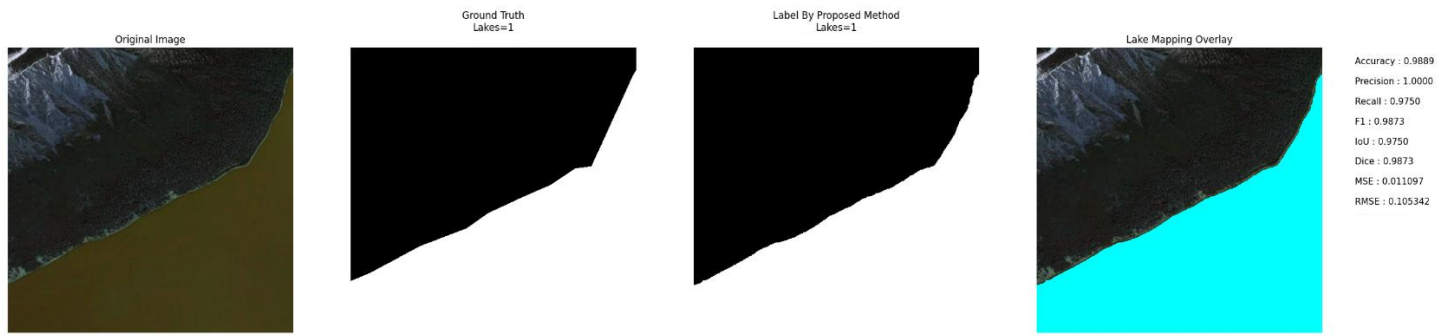


Image No: 306

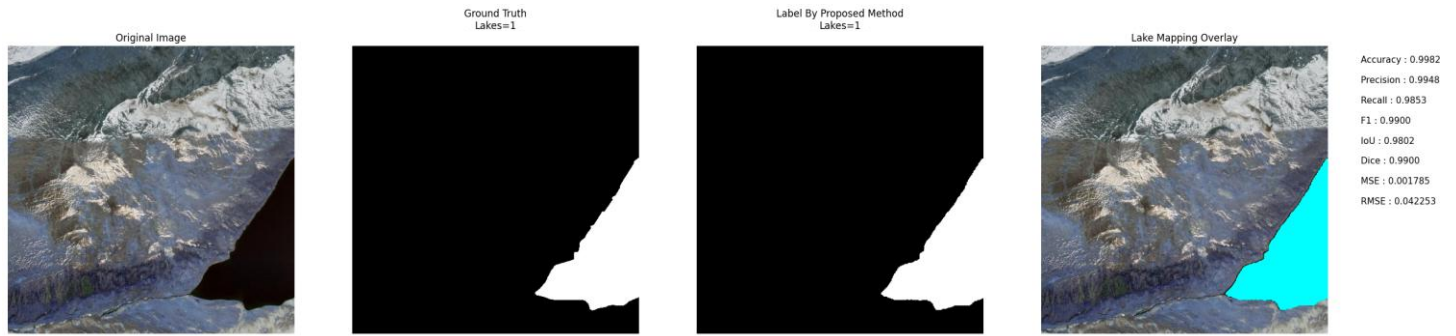


Image No: 850

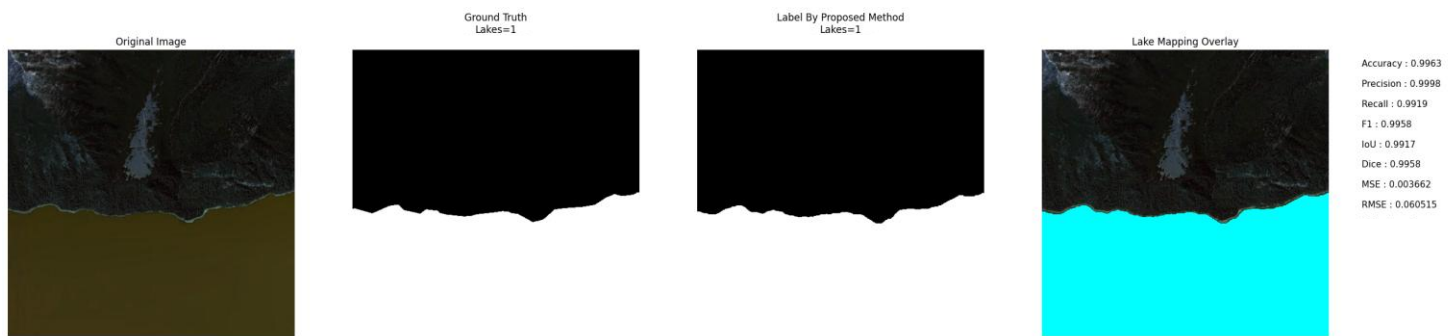


Image No: 774

4. Debris Cover

Description: Output images obtained for debris-covered regions.

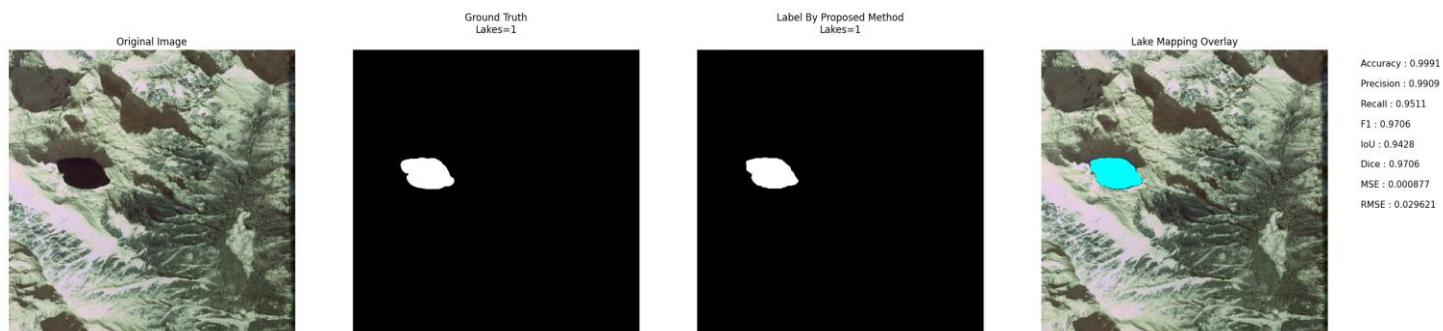


Image No: 529

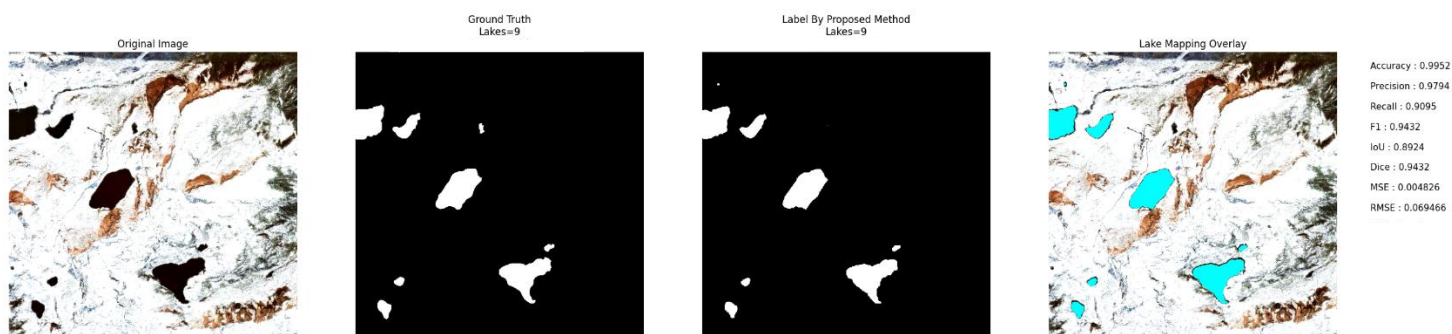


Image No: 41

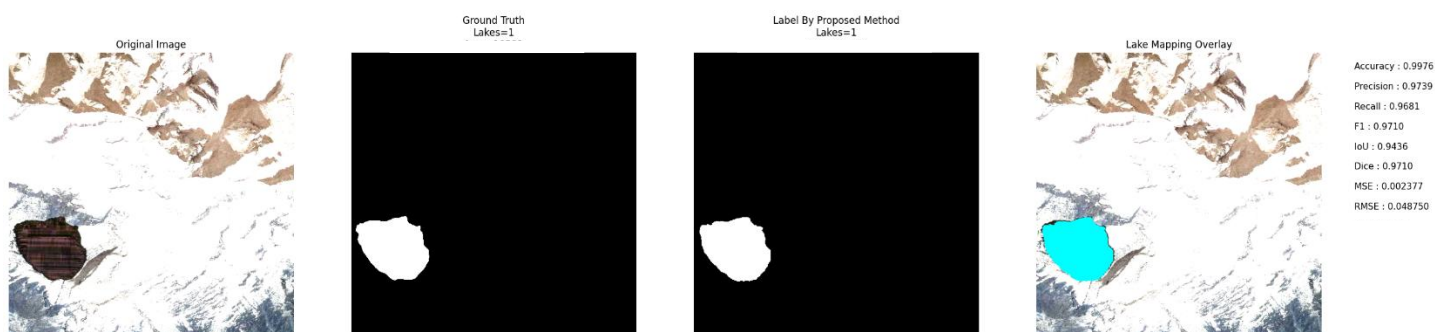


Image No: 176

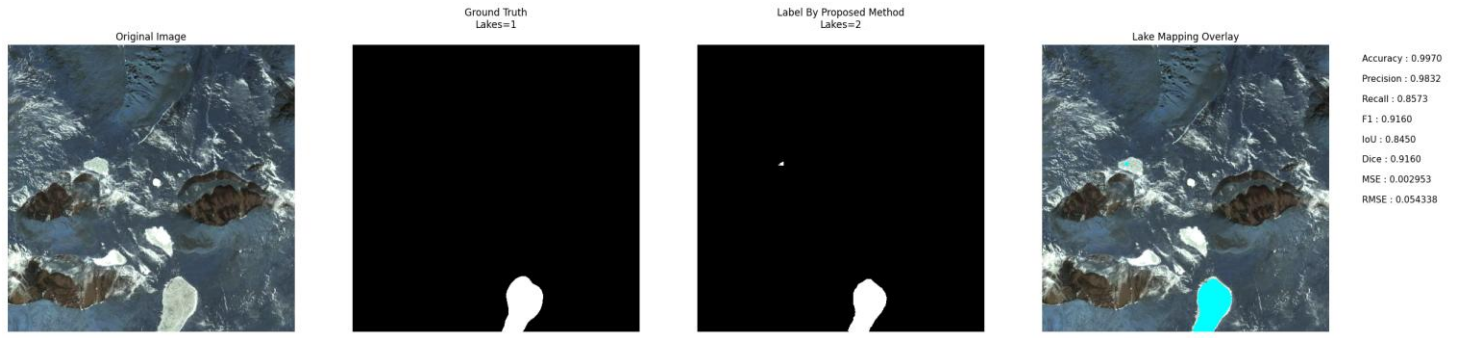


Image No: 575

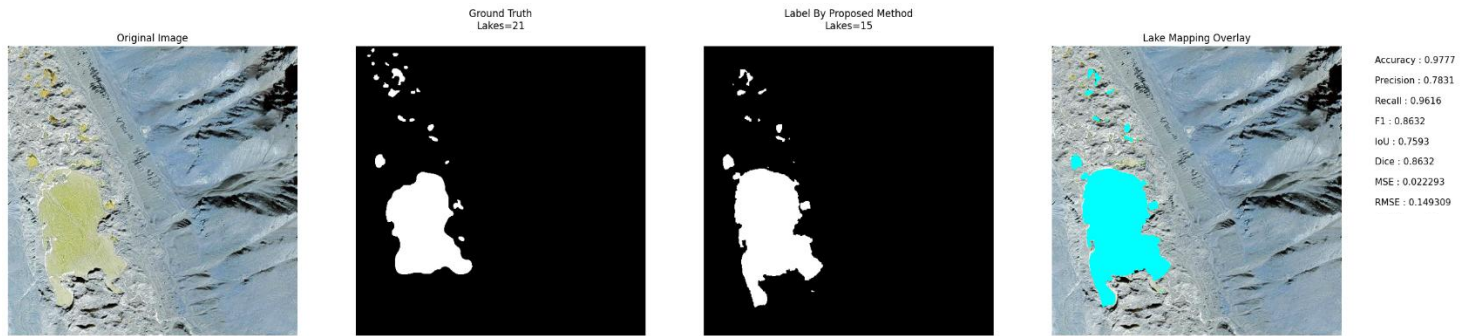


Image No: 113

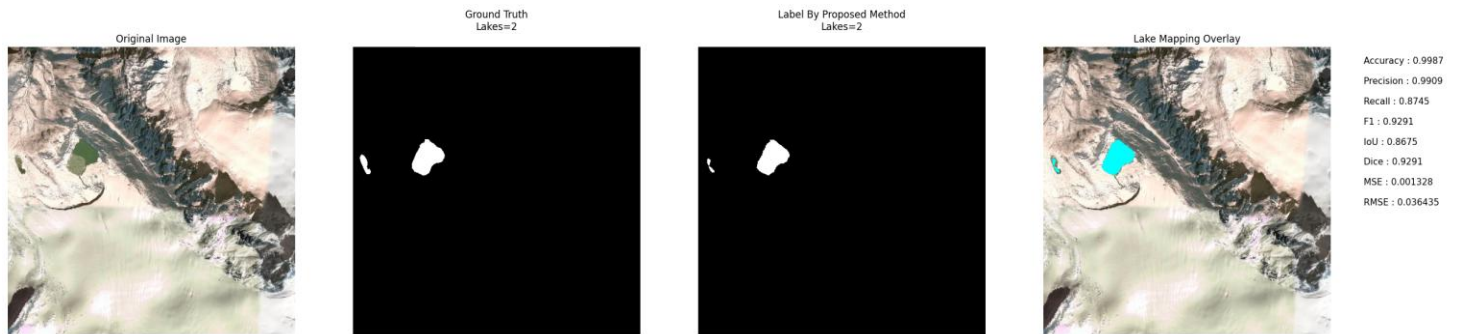


Image No: 448

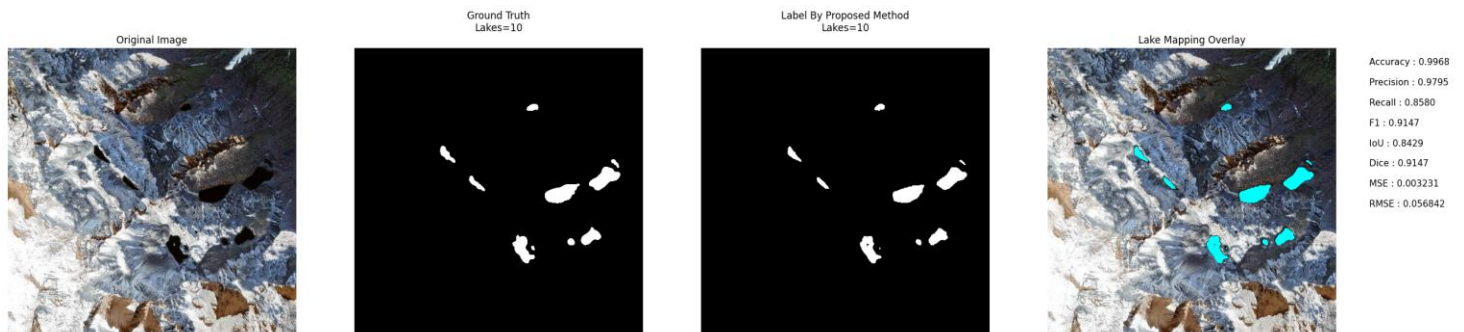


Image No: 972

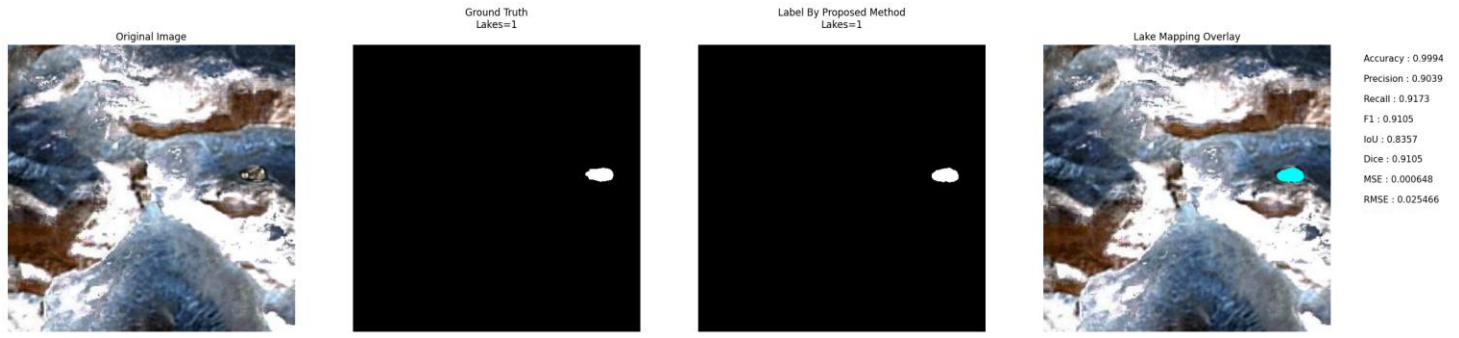


Image No: 951

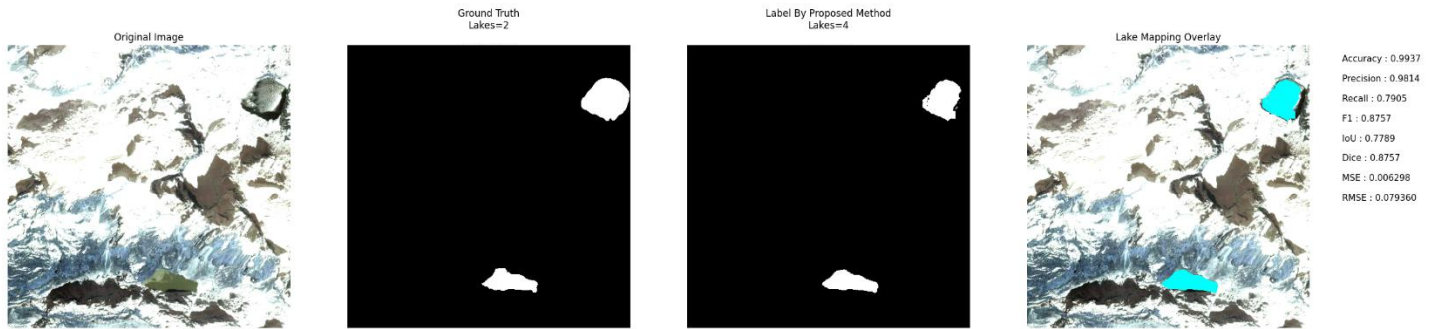


Image No: 1895

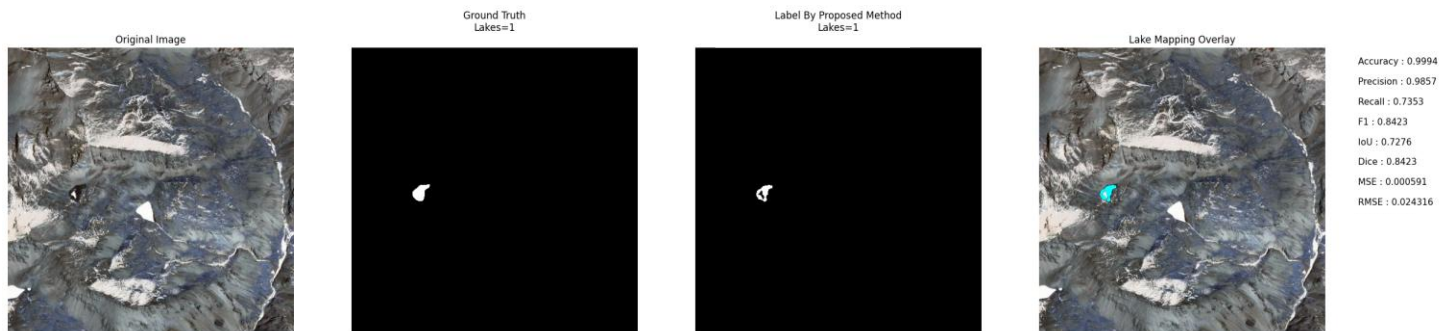


Image No: 1600

5. Moraine-Dammed Lakes

Description: Output images of moraine-dammed glacial lakes.

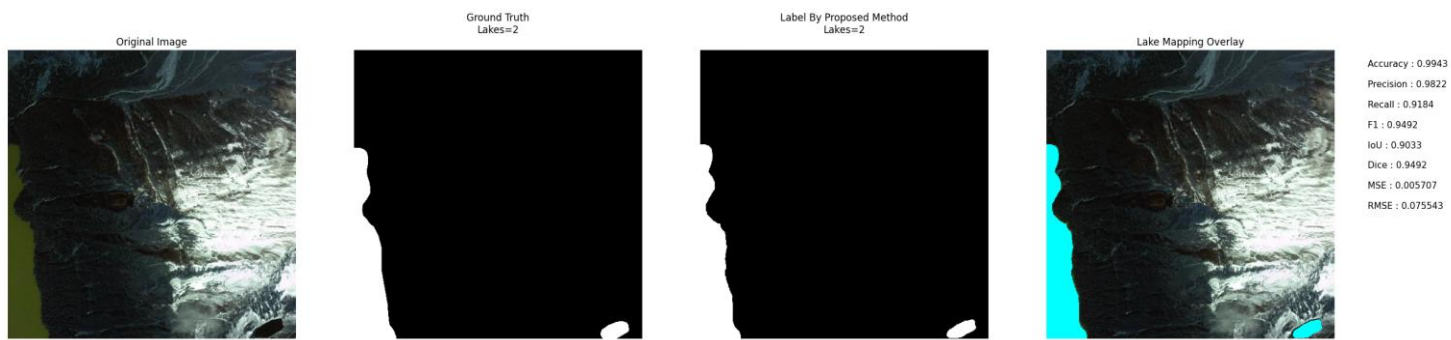


Image No: 5

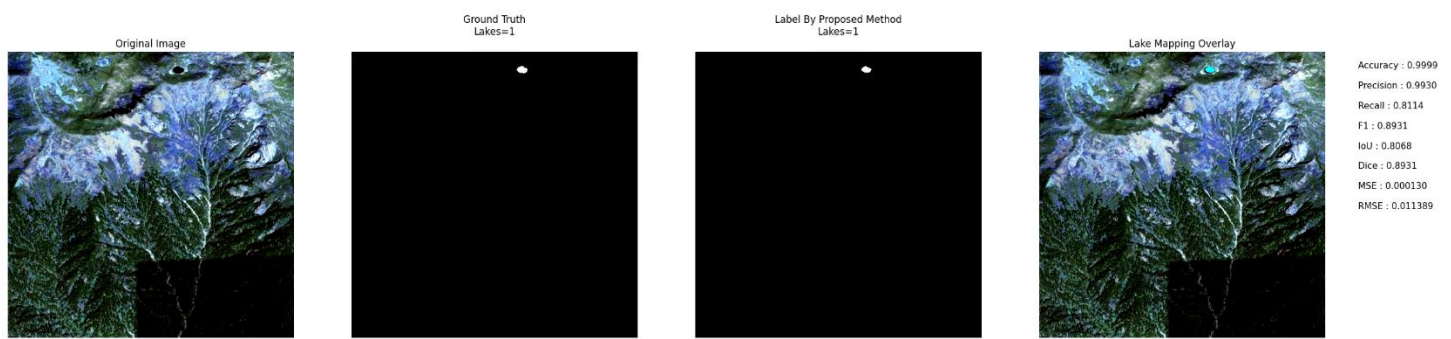


Image No: 15

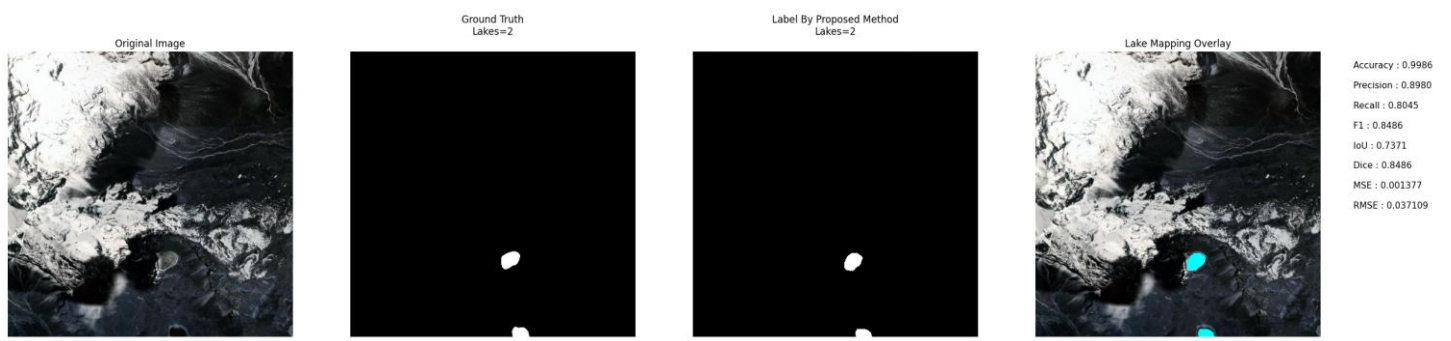


Image No: 17

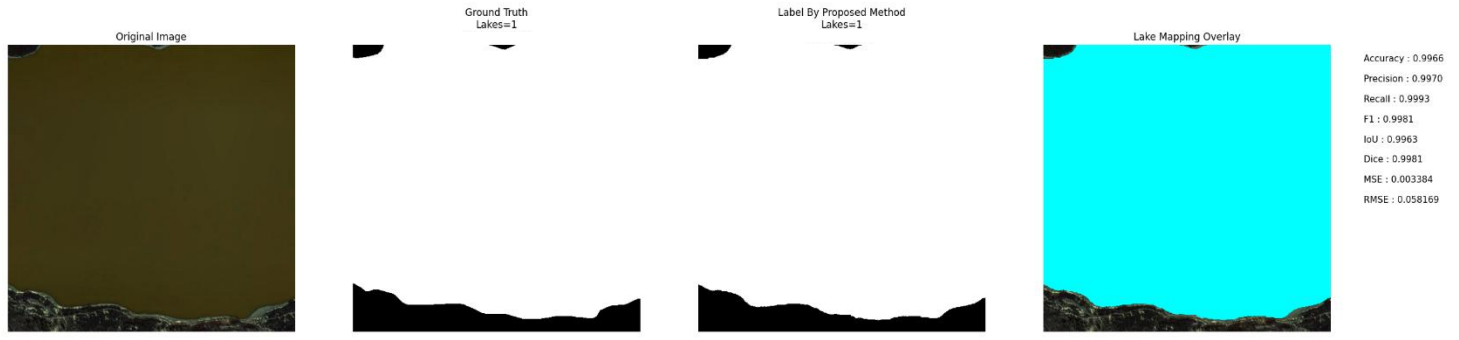


Image No: 20

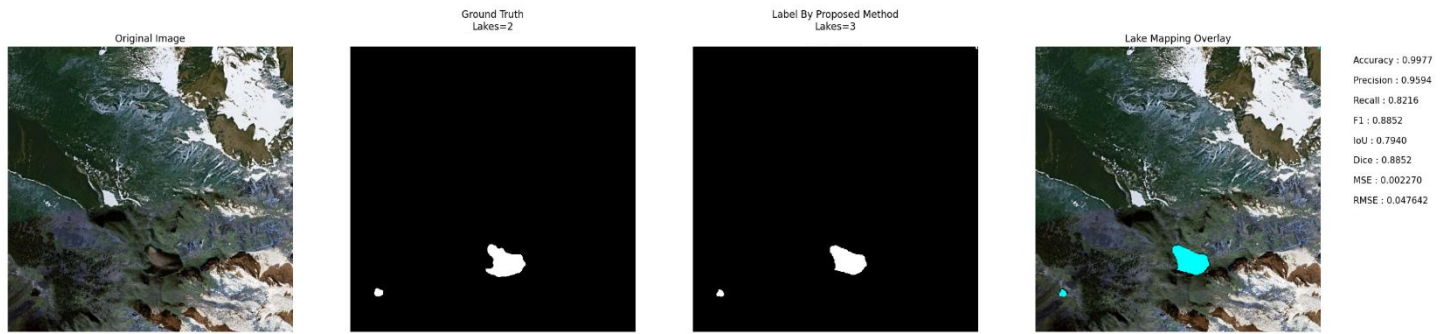


Image No: 31

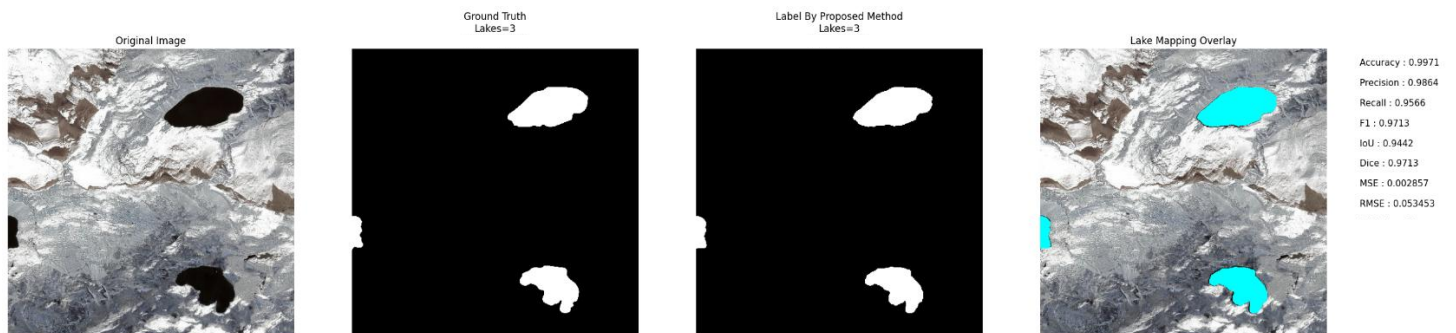


Image No: 557

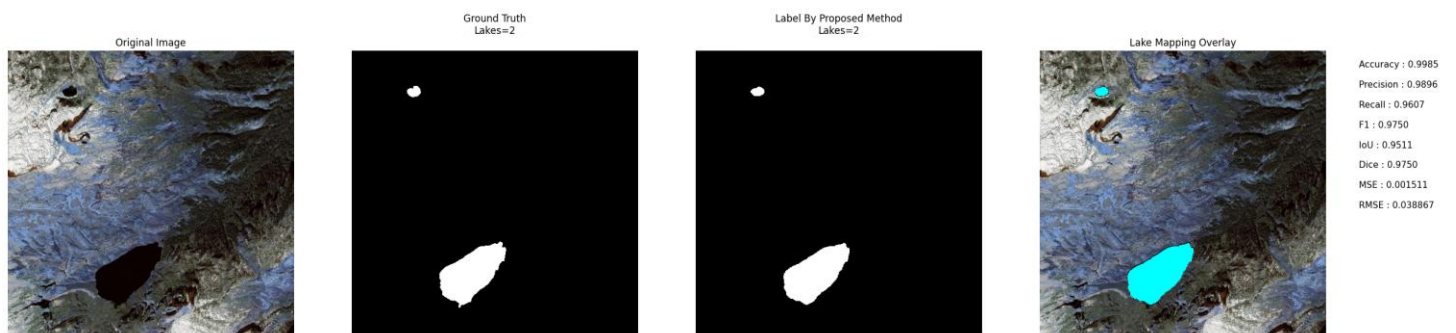


Image No: 65



Image No: 228

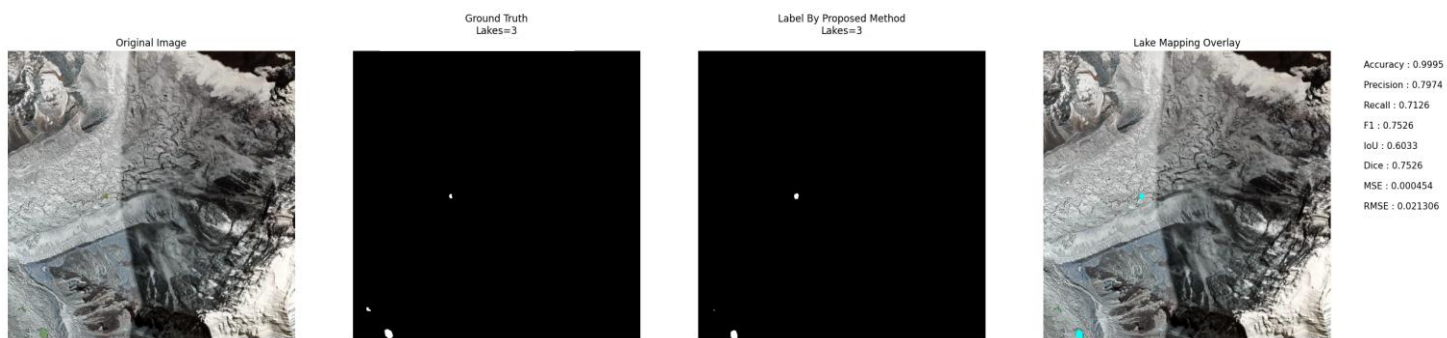


Image No: 229

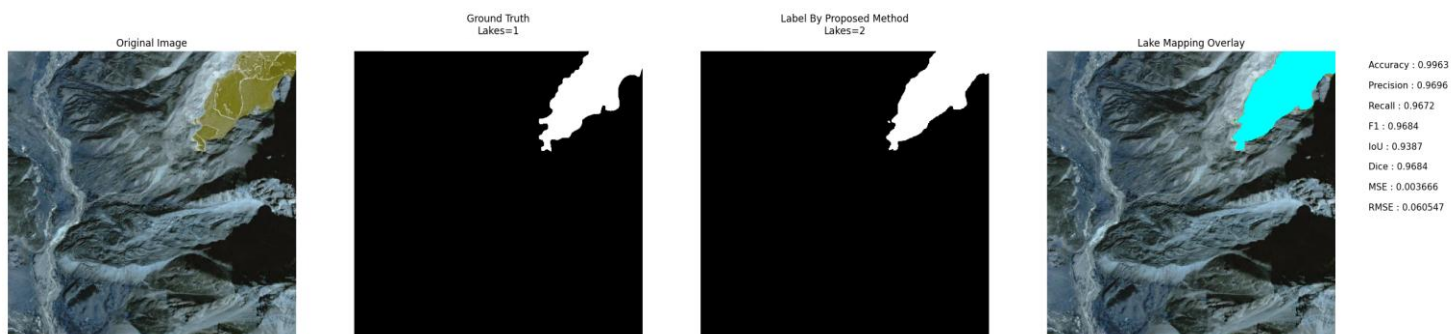


Image No: 50

6. Varying Turbidity

Description: Output images containing lakes with different turbidity levels.

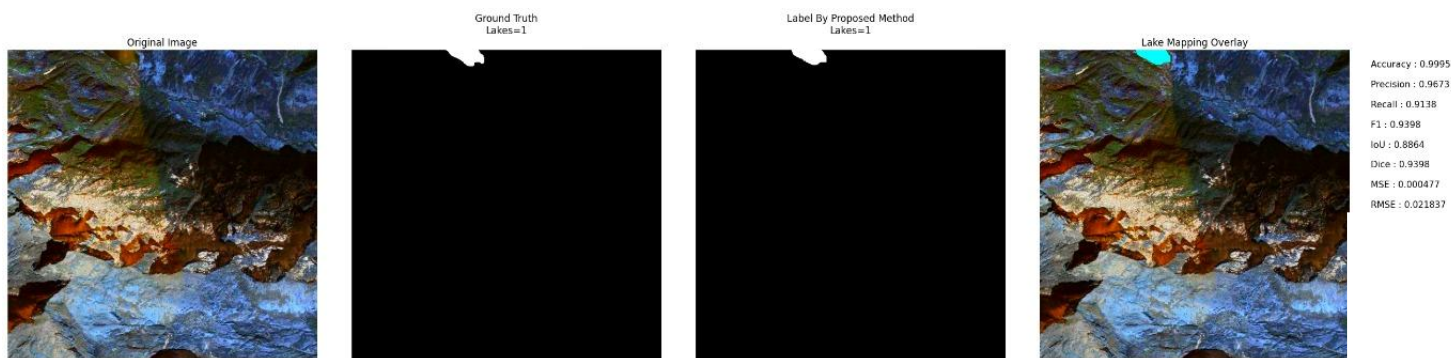


Image No: 1769

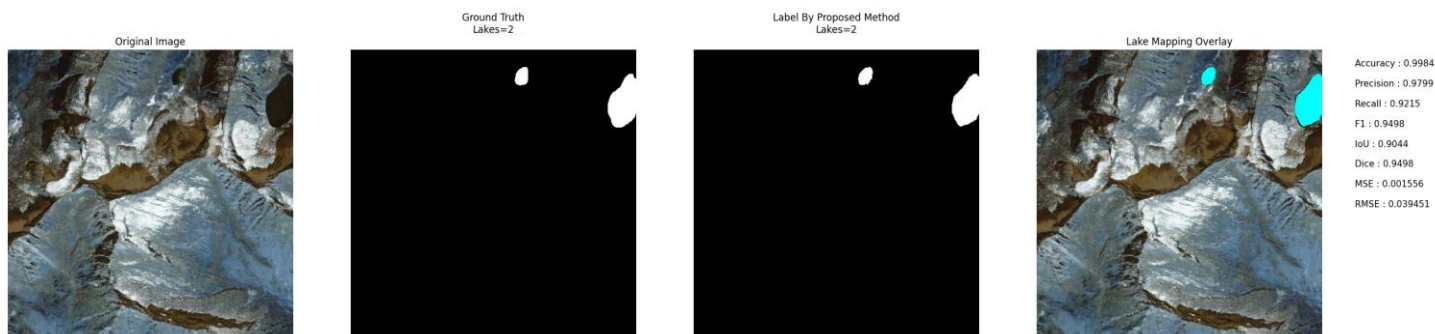


Image No: 1297

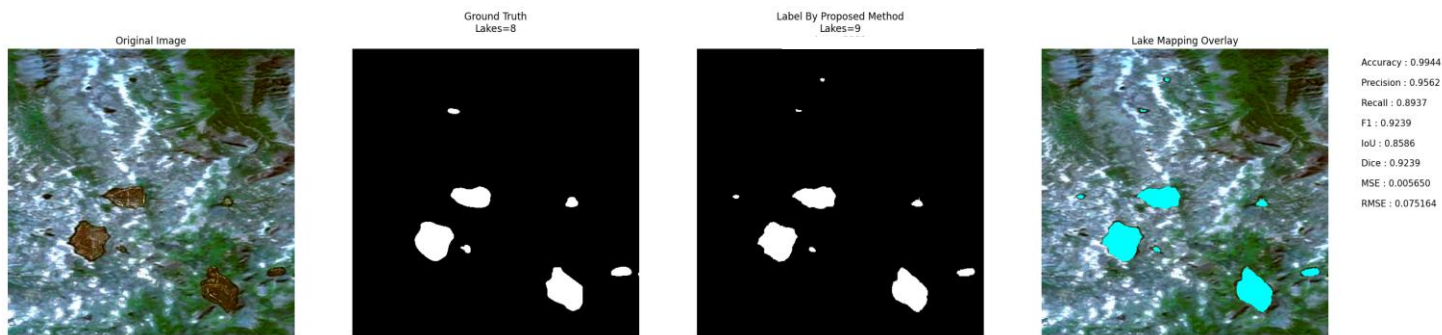


Image No: 1354

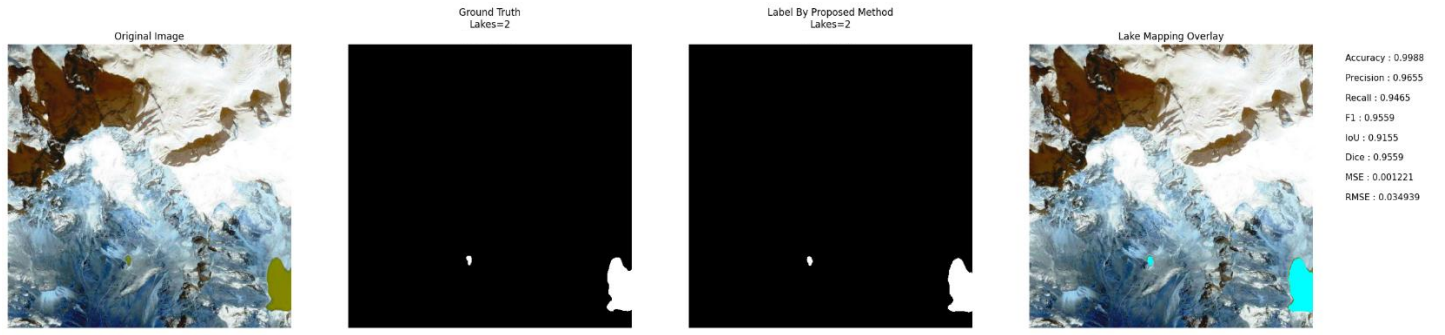


Image No: 1976

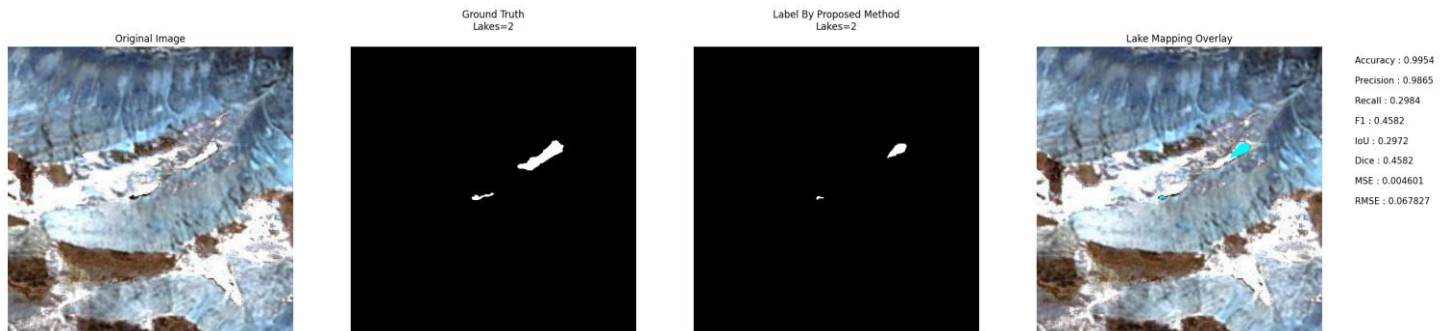


Image No: 1651

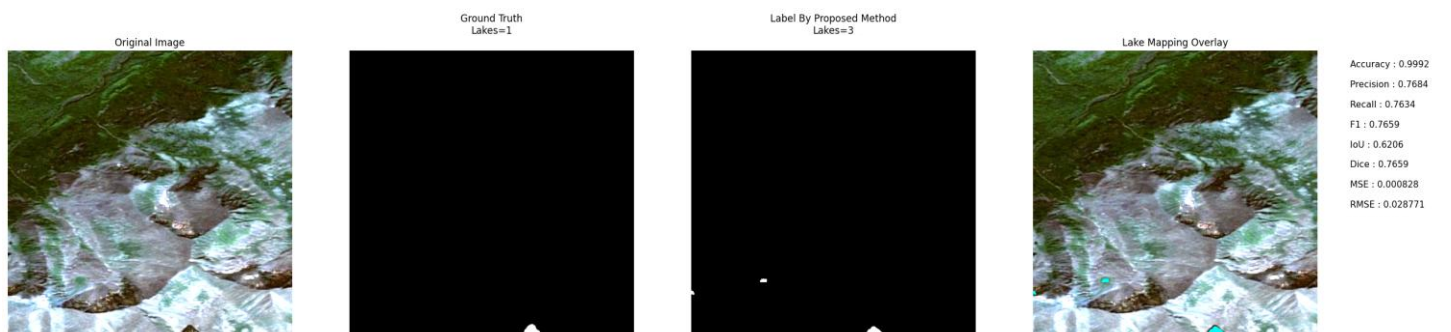


Image No: 1789

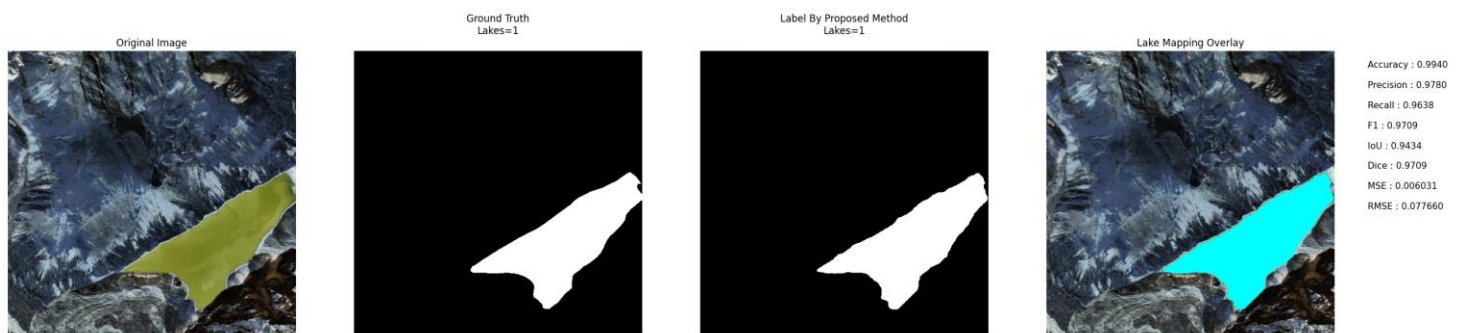


Image No: 1823

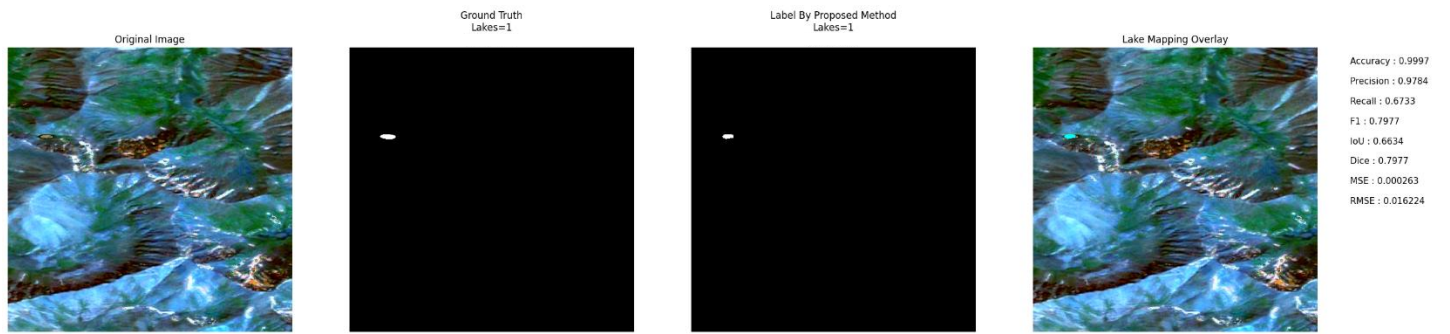


Image No: 1686

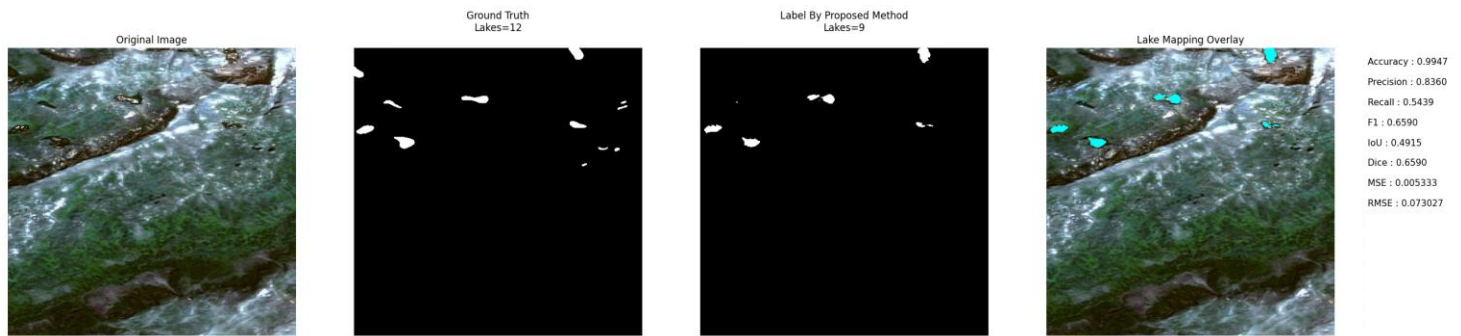


Image No: 2208

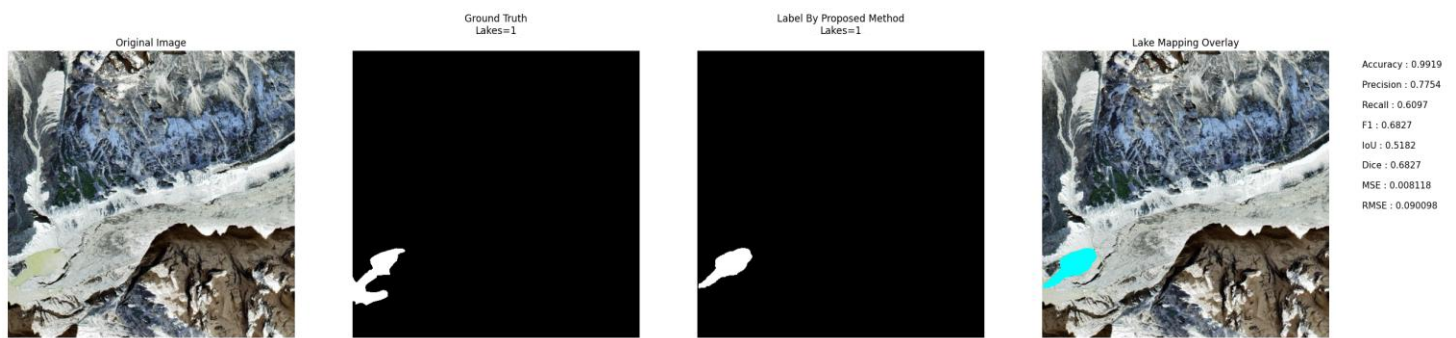


Image No: 1882

Conclusion

The generated outputs demonstrate the effectiveness of the proposed framework in detecting glacial lakes under various challenging environmental conditions, including cloud cover, snow cover, terrain shadows, debris-covered regions, moraine-dammed lakes, and varying turbidity levels. The model exhibited strong generalization capability across diverse glacial landscapes and successfully differentiated glacial lakes from visually similar features such as snow, shadows, and debris-covered surfaces. Validation results indicate a high detection accuracy

of **97.96%** with a low **Mean Squared Error (MSE) of 0.0165**, highlighting the reliability and precision of the proposed approach. The consistent performance across all six classes demonstrates the robustness of the framework for large-scale glacial lake mapping and monitoring. These results suggest that the proposed model can serve as a valuable tool for Glacial Lake Outburst Flood (GLOF) risk assessment, environmental monitoring, and climate change adaptation strategies.

Due to space limitations in the report, a representative set of 10 output images from each class has been included for visual analysis. The complete set of output images generated during the experimentation phase, including additional results for Cloud Cover, Snow Cover, Terrain Shadows, Debris Cover, Moraine-Dammed Lakes, and Varying Turbidity classes, is available through the following Google Drive repository:

Google Drive Link:

https://drive.google.com/drive/folders/1JtTRxqoB_1j_qsyPzL5_2HdQOv-Hg7eh?usp=sharing

The repository contains all output images used for model validation and qualitative performance assessment.